SHREYAS RAORANE

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Education

UNIVERSITY OF PENNSYLVANIA

Master's in Robotics, GPA: 3.8/4.0 *Aug 2024 – May 2026*

Work Experience

EVOSORT SIMULATION *Jun 2025 – Aug 2025*

OPEX Corporation, Time Critical Software Engineering Intern

- Engineered multi-threaded discrete-event simulation in C++ for warehouse automation, managing 1,000+ parallel material flow events with shared state across 3 concurrent threads
- Developed a time-critical scheduling engine completing actuation sequences within 3 ms. This modelled hardware timing for production validation of the software
- Refactored legacy codebase with shared header architecture and a centralized `SimulationHandler`, letting the software team validate against the simulator instead of waiting for limited physical hardware availability

SOFTWARE DEVELOPMENT (ESAP Enterprise Platform) *Jun 2022 – Jun 2024*

Brillio Technologies, Software Development Engineer

- Optimized 40+ PostgreSQL queries to cut response time 35%; as sole developer on the Wholesale EMEA module of a Verizon enterprise platform (10,000+ users), built and owned 20+ production screens end-to-end
- Placed 3rd in Brillio's internal Hack-o-Holics hackathon, building a web app that transcribes and summarizes meetings into auto-generated minutes via a speech-to-text model

Robotics Projects

AUTONOMOUS DRIVING STACK (F1/10 Platform) *Feb 2025 – Mar 2025*

University of Pennsylvania *[Code]*

- Implemented reactive planners (wall-follow, gap-follow, pure-pursuit) and an RViz waypoint mapping/logging pipeline in ROS2, integrating particle-filter localization and SLAM modules into a LIDAR + IMU autonomy stack
- Bridged the AutoDRIVE Simulator and F1/10 hardware over a shared interface with parameter tuning for sim-to-real transfer; added an RGB-D camera feed for richer scene visibility

MASTER'S THESIS: SCALABLE MULTI-ROBOT COORDINATION (GRASP Lab) *Oct 2025 – May 2026*

Advisors: Prof. Phan, Prof. Mangharam, Prof. Hsieh *[Code]*

- Designed LCBA (Local Consensus-based Bundle Algorithm), a decentralized task allocation algorithm extending GCBBA to intermittently disconnected, dynamic communication graphs via component-local consensus and event-driven replanning
- Integrated LCBA with three interchangeable planners (Cooperative A*, PBS, RHCR); across five maps (6–200 agents, 432+ runs) against CBBA, SGA, and DMCHBA, sustained ~6x higher throughput than CBBA at high load within the real-time budget
- Maintained non-zero task completion in scenarios where CBBA and SGA baselines dropped to zero

SBAMP: SAMPLING-BASED ADAPTIVE MOTION PLANNING *Mar 2025 – Apr 2025*

University of Pennsylvania *[Code]*

- Built adaptive RRT* planner with dynamical-systems attractors for real-time goal tracking; replanned fast enough to track goals where static RRT* could not keep up
- **Published at the ICRA 2026 Workshop on Frontiers of Optimization for Robotics —** [Paper]

SwarmSync: DISTRIBUTED MPC FOR QUADROTOR SWARMS *Oct 2025 – Dec 2025*

University of Pennsylvania *[Code]*

- Extended the online DMPC framework (Luis et al.) with moving goals and an online Hungarian task-reallocation layer that re-assigns drones to goals to minimize total travel distance
- Generated collision-free trajectories via per-agent QP optimization; scaled to 32-drone demos and a 64-agent sweep with zero collisions and 100% goal success across all 9 scenarios at $>$200 Hz solve rate

QUATERNION-BASED UKF FOR ORIENTATION TRACKING *Sep 2024 – Dec 2024*

University of Pennsylvania *[Code]*

- Implemented quaternion-based Unscented Kalman Filter (UKF) sensor fusion of 3-axis gyroscope and accelerometer IMU data for drift-free 3D orientation estimation
- Propagated sigma points on the quaternion manifold and validated roll, pitch, and yaw against Vicon ground truth

Skills

Programming: C++, Python, MATLAB, CUDA, Object-Oriented Design, Design Patterns

Systems: Multi-threading, Real-time Systems, Memory Management, Distributed Systems, Event-driven Architecture, Linux

Perception & Localization: Perception, Sensor Fusion (EKF, UKF, Particle Filter), SLAM, Localization & Mapping, Obstacle Detection, Object Detection (YOLO), Sensor Integration (LIDAR, Cameras, IMU, RGB-D)

Motion Planning: RRT*, A*, MPC, DMPC, QP Optimization, Sampling-Based Planning, Collision Avoidance

Robotics & Tools: ROS2, Multi-Agent Coordination, IsaacSim, Gazebo, PyBullet, AutoDRIVE, Foxglove, Docker, Git, CI/CD